

Endocrinopathies in Exotic Companion Mammals

AKA “Greater than 90% of Ferret Medicine...”

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Learning Outcomes

1. **Compare expected clinical presentations of the most common endocrine diseases that affecting exotic companion mammals.**
2. **Differentiate endocrine diseases from other differential diagnoses using species-specific clinical findings and diagnostic test results.**
3. **Develop evidence-based treatment plans for exotic companion mammals diagnosed with common endocrinopathies.**

Endocrinopathies are common differentials for disease presentation, particularly in ferrets. Although the basic endocrine physiology is the same amongst mammals, disease presentation can be very different from traditional small animals (dogs and cats) requiring a separate knowledge base to effectively diagnose and treat these conditions.

Diseases of the hypothalamic-pituitary-adrenal axis

Ferret Adrenocortical Disease (ACD)

Pathophysiology: The most common form of ferret ACD and one of the most common diagnoses of ferrets is characterized by hyperandrogenemia. This endocrinopathy is caused by hypertrophy/hyperplasia (most common), adenomas, or adenocarcinomas of the zona reticularis of the adrenal cortex. Several theories have been considered spanning from effects of early neutering to rearing under artificial lighting to genetics.

Clinical Presentation: ACD is typically seen in middle to older aged ferrets without a sex predilection. The most common clinical sign is progressive alopecia (90% of cases) and at least a third of ferrets will also present with pruritis, leading to excoriations and visible scratches. Jills often present with vulvar swelling that may have an associated vaginitis. Affected hobs are at risk for presenting with lower urinary tract straining or obstruction secondary to prostatomegaly or prostatic cysts, which may become secondarily infected. Urinary obstructed ferrets may present emergently with metabolic derangements and a similar presentation to a feline urinary obstruction cat. In a small proportion of cases, the adrenal gland is palpable on palpation, however, this should not be a criterion for diagnosis.

Differential Diagnoses:

- Pruritic hair loss can be caused by ectoparasites, dermatophytosis, and environmental allergens (rare).
- Vulvar enlargement may be associated with cycling intact female, remnant ovarian syndrome, or exogenous estrogen exposure.
- Lower urinary obstruction may also be caused by urolithiasis or lower urinary neoplasia.

Diagnosis:

- Complete blood cell count is often unremarkable, however in rare cases of hyperestrogenism, pancytopenia may be seen.
- Serum/plasma chemistry is often unremarkable.
- As most adrenal gland enlargement is mild to moderate, radiographs are often unremarkable.
- Abdominal ultrasound is one of the diagnostic criteria for ACD detection. Adrenal gland measurements can be compared to published species-specific measurements. The biggest advantage of ultrasound is in the detection of other comorbidities, which are common in ferrets (prostatomegaly in males, insulinoma, enlarged lymph nodes, gastrointestinal thickening)
- There is a ferret specific adrenal panel available that measures the three most common androgens to be elevated (17-alpha hydroxyprogesterone, estradiol, and/or androstenedione). This panel can be a useful tool when ultrasound is not available for diagnosis and may help to monitor response to therapy.

Treatment: There are both medical and surgical management options available for the management of the ferret diagnosed with ACD. The method of management chosen will depend on clinician preference, patient health status and comorbidities, and the presence of vascular invasion of the adrenal enlargement. The goal of medical management is to manage the clinical signs by reducing androgen concentrations, but cannot “cure” the disease, as the adrenal tumor will not decrease in size or rate of growth. Deslorelin (Suprelorin) is a long-acting implant (size of a microchip) that releases a GnRH analogue and is labelled and licensed for use in ferrets with ACD. The mechanism of action causes initial stimulation of the hypothalamic-pituitary axis and then eventual suppression (by 2 weeks after placement) of the axis due to a feedback inhibition loop. During these 2 weeks, clinical signs may worsen which may be particularly dangerous for male ferrets as they may present with life threatening prostatomegaly. This risk may be mitigated by the effects of flutamide (androgen receptor antagonist). The major advantages of surgical intervention are the ability to remove the source of the problem (ie “cure” the disease) and to address other diagnoses under the same anesthetic event (ex: partial pancreatectomy). While left adrenalectomy is rarely complicated, the right adrenal gland is found dorsal to the vena cava and requires diligent dissection. Unilateral adrenalectomy may be complicated by later development of ACD in the contralateral adrenal gland. Removal of both adrenal glands can be performed, however often require corticosteroid and mineralocorticoid supplementation.

Prognosis: In most cases, the adrenal mass is benign, and ferrets can be managed with this diagnosis, even with the detection of other comorbidities for several years. Factors that reduce the prognosis are the presence of a right adrenal gland that invades the caudal vena cava and/or the development of lower urinary tract obstruction in hobs.

Other notes: Ferret ACD is NOT Cushing’s disease. Although both diseases are adrenocortical endocrinopathies, they have different presentations, diagnostic criteria, and treatment considerations. #notCushing’s

Rabbit Adrenal Tumors

Although uncommonly diagnosed, rabbits develop also adrenal tumors that produce androgens. The most common complaint is a spayed or neutered rabbit that develops sexual activity and/or urinary spraying later in life. Adrenal adenomas and adenocarcinomas are reported and both medical and surgical management are reported, however, are limited to case reports currently.

Syrian Hamster Adrenocortical Diseases

Laboratory hamsters have a high incidence of adrenocortical hyperplasia and adenomas. Infrequently, Cushingoid Syrian hamsters may be seen that exhibit polyuria, polydipsia, truncal alopecia, and hyperpigmentation. Although the small size of the species makes diagnosis challenging, serum cortisol levels can confirm the diagnosis and a variety of medical management options have been described with varying success.

Diseases of the hypothalamic-pituitary-thyroid axis

Guinea Pig Hyperthyroidism

Pathophysiology: Hyperthyroidism can be a consequence of thyroid hyperplasia or neoplasia (benign and malignant reported). Thyroid tumors can also be nonfunctional. Hyperthyroidism is the most common endocrinopathy of guinea pigs.

Clinical Presentation: Affected guinea pigs may present with some of the classic effects of increased thyroid hormones noted in traditional species: weight loss and reduced body condition despite an adequate or increased appetite, polyuria and polydipsia, and hyperactivity. Physical examination findings that may support the diagnosis include the detection of hyperthermia and tachycardia or tachyarrhythmias. Many guinea pigs may have low grade diarrhea (soft stools) that may act as a red herring. Ventral cervical palpation is an important part of the physical examination of guinea pigs to screen for subcutaneous masses.

Differential Diagnoses:

- Ventral cervical masses may also be an abscess or granuloma, lymphadenopathy (lymphoma or cervical lymphadenitis) or sialoceole.
- Primary polydipsia and secondary polyuria is often seen in individually housed guinea pigs, or those that recently lost a cage mate. It can also be seen in states of chronic renal failure.
- Given that many clinical presentations may present with weight loss, a thorough history, physical exam and consideration of minimum database should be performed in these cases.

Diagnosis:

- Complete blood cell counts and serum/plasma chemistry is often unremarkable.
- Thyroid concentrations can be measured, however, specialized labs that use radioimmunoassay (gold standard) should be used. Often these labs may require large plasma volumes (>1mL) that are clinically challenging to obtain.
- Although radiographs are rarely useful, ultrasound and computed tomography may anatomically relate the cervical mass to other structures and guide fine needle aspiration.
- Cytologic evaluation of a fine needle aspirate is an important component of the diagnostic criteria, especially if the adequate volume of blood cannot be collected for thyroid hormone concentration measurement.

Treatment: Surgical (thyroidectomy), medical (methimazole) and radiological (I-131) therapy have all been reported in the management of hyperthyroidism in guinea pigs. Cost and availability as well as patient factors may direct recommendations.

Other notes: Guinea pigs also get hypothyroidism, but much less commonly!

Ferret Thyroid Disorders

Although rare, hypothyroidism has been reported in ferrets. Interestingly, the clinical findings classically seen in dogs (hair loss, hypoglycemia, and hypercholesterolemia) have not been reported in ferrets. Several thyroid neoplasia have been reported in ferrets, however, no functional thyroid neoplasia have been reported.

Diseases of the Endocrine Pancreas

Ferret Insulinoma

Pathophysiology: Pancreatic islet cell tumors arising from the beta cells of the pancreas and creating excessive insulin are one of the most common diagnoses in ferrets. Although most tumors are benign, the effects of increased insulin (hypoglycemia) can be life threatening. Several underlying causes have been postulated including genetics and unnatural diets.

Clinical Presentation: Ferrets with insulinoma may present with signs of acute or chronic neuroglycopenia. In acute neuroglycopenia, the presentation is often emergent as the ferret may be suffering from seizures or present minimally responsive. States of chronic neuroglycopenia may be more insidious in nature including incoordination and lethargy. In many cases, ferrets will exhibit nausea as evidence of them drooling and/or pawing at their mouth when they present. Clients may share the signs are intermittent, although they may not be able to relate them with the recent ingestion of food.

Differential Diagnoses:

- Seizures may be seen with some toxicities (acetaminophen), infection with canine distemper, traumatic brain injury, or an intracranial lesion.
- Drooling and pawing at the mouth may also be caused by other metabolic presentations (chronic renal failure) or primary upper gastrointestinal presentations (gastritis, gastric foreign bodies, enteritis).
- Although the most common cause of hypoglycemia in ferrets is insulinoma, other causes should be considered: artifactual, sepsis, and liver failure.
- Due to the large number of conditions that may cause lethargy, a full a thorough history, physical exam and consideration of minimum database should be performed in these cases.

Diagnosis:

- Complete blood cell count and serum/plasma chemistry is often unremarkable outside of hypoglycemia. Because of the normally high hematocrit of ferrets, hypoglycemia is often noted on chemistry panels where the blood is not analyzed immediately.
- Patient side blood glucose measurement is the most important screening tool. The Alpatrak veterinary glucometer on the canine setting is validated for use in ferrets and should be evaluated in every sick ferret that presents. It is important to remember that a low blood sugar documents hypoglycemia, and the most common cause of hypoglycemia is insulinoma.
- Measurement of insulin concentrations (or glucose:insulin) may be helpful. In normal states, hypoglycemic patients should have low insulin concentrations.
- While abdominal ultrasound and computed tomography can document pancreatic nodules, abdominal radiographs lack sensitivity for detecting lesions

Treatment: The emergency management of the acutely affected ferret should focus on arresting seizure activity with benzodiazepines while increasing blood sugar through intravenous administration of dextrose. Injectable administration of dextrose should be reserved for ferrets that cannot swallow food on their own, as providing a high protein meal is more appropriate in cases of chronic neuroglycopenia. Surgical (partial pancreatectomy or nodulectomy) and medical management options are available. The hallmark of medical management is steroid administration, however diazoxide and octreotide acetate are also used. Both surgical and medical management require life long monitoring of blood sugar.

Other notes: Unlike dogs, where insulinoma has a high rate of metastasis at the time of diagnosis, ferret insulinomas are typically benign.

Guinea Pig Insulinoma

Although uncommonly diagnosed, older guinea pigs may present with acute neuroglycopenia and acutely lethargic guinea pigs and those with neurologic signs should have their blood sugar checked in clinical settings. Although most cases were diagnosed on postmortem, there is documentation of clinical management with diazoxide.